

YOUR OWN SUPERCOMPUTER

With Super Cool Tech

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There are areas where we need small computing machines that can work collectively like a supercomputer. Let us see here how it can be done.

Most supercomputers have inter-

connected CPUs with a master node CPU that divides a large computing task into smaller tasks. These smaller tasks are assigned to the interconnected CPUs, which work together to finish the task in much

lesser time.

So let us design our own supercomputer, which would be a basic system where you can connect as many nodes as you want to fulfill your computation and processing need. We shall use single-board computers and connect them to each other and make one of them the master node. The master node would distribute the tasks and control all the other computers.

The components required for this project are listed under the Bill of Material table. The author's prototype is shown in Fig. 1.

Each node of the cluster for our supercomputer would need the com-

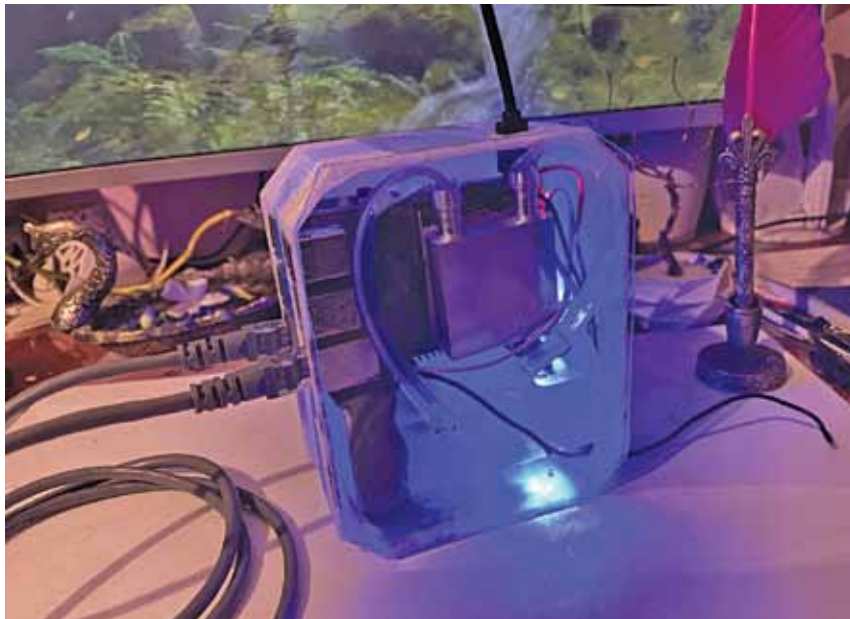


Fig. 1: Author's prototype

BILL OF MATERIAL

Components	Quantity	Description
Raspberry Pi 4GB	Min 2 nodes	Supercomputer nodes
Power adaptor	Min 2 per node	2A
Water-cooled aluminium block	Min 2 per node	40mm×40mm×12mm
SD card 16GB/SSD 120GB to 500GB	SD card min 2 / 1 SSD	100MB/s R/W speed
Thermoelectric cooler	Min 2 per node	40mm×40mm×12mm
7mm dia pipe	Min 4 per node	7mm 5mm orifice
Water or cooling liquid	200ml	Water/Cooling fluid
Water reservoir	200ml	Cylindrical air-tight container
Micro water pump	1 per node	3-5 V 3mm orifice
Ethernet cable	2 per node	15cm Ethernet cable
3D printed case	1 per node	PLA/ABS 3D printed
Ethernet splitter	1	2 or more ports

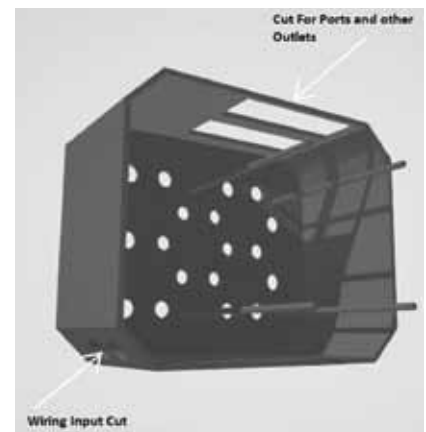


Fig. 2: The case design for a block



Fig. 3: Inlet and outlet pipes fixed in water reservoir